

SAI SUCHITH D

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PROFESSIONAL SUMMARY

6 years of experience as a Data Engineer with concentration in implementing scalable data pipelines and ETL architecture. Adept in **Azure** and **AWS** eco-system for data log processing and anomaly detection.

- Immense knowledge of **Snowflake data platform** with **Snowflake** Multi-cluster and Snow Pipe along with **Hadoop, Spark** for better business strategies and enhance overall data management and performance using computing frameworks.
- Incorporated and deployed **ETL** migration services using **AWS Lambda** for generating serverless **data pipelines** which are queried using **Athena** and can be written to **Glue Catalog** and use of **S3, Redshift, EMR, Glue and airflow** for big data to check system logs and identify potential issues.
- Use of Amazon Web Services like **S3, EC2, REDSHIFT, REDIS, EMR, Kinesis, Lambda, Step Functions, Terraform, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch and IAM, EDS, Cloud Run, Cloud Functions** to store structured data, make necessary transformations, simplify data integration, run SQL queries and run ETL to facilitate smooth data analytics.
- Expertise in use of **Microsoft Azure** services like **Azure Data Lake, Data factory, Azure Databricks, Azure SQL database, Azure SQL Datawarehouse, Azure Synapse Analytics, Azure Data Lake Storage, Azure Stream Analytics, Azure Cosmos DB, Azure Machine Learning, Azure Databricks, Azure PostgreSQL** to move data from sources, creating visual workflows and providing flexibility of data which is suitable for transactional applications, it is also used to build, manage and optimize data pipelines and analytics which is helpful for data processing and optimize data pipelines.
- **PySpark** is used for developing **ETL applications** for large volumes of data and extracting this data to perform SQL queries. Use of **Microsoft Fabric** to design pipeline and perform scalable solutions across enterprise data platforms.
- Automated process in **Azure** to extract data from web services and load into **SQL database**. **Azure Data Factory** pipelines are used to move data from on-premises to **Azure SQL Warehouse** and from **Amazon S3** buckets to **Azure blob storage** to manage high-scale workloads and support real-time batch processing.
- Engineered **ETL pipelines** by using **Python, Scala, and Java**, utilized **Spark Streaming, and Flink** to manage real-time data processing and increasing throughput using **parallelization, vectorized execution, and batch processing** techniques.
- Expertise in **SQL, PL/SQL, Hive, Oracle, MySQL, PostgreSQL, and DB2**. Also writing **complex, multi-join queries** that leverage **indexing, partition pruning, materialized views, and query caching** for query optimization to enhance performance, reduce execution time for high-volume databases.
- For big-data applications **Apache Spark** is used for distributed data processing, **Spark SQL** for data query, **Apache Sqoop** for data migration, **Apache hive** for data warehousing and data processing like collecting, aggregating and moving data from various sources using **Apache Fume and Kafka** to ensure data availability.
- **ETL workflows** optimized using **Apache Airflow, Oozie, Azkaban and AWS Step Functions** to implement **DAG** scheduling and SLA monitoring and to adapt **serverless ETL workflows** for cost-effective data pipeline execution.
- Address data lake solutions using **Databricks** for efficient data driven solutions, unified data analytics, ETL, integrating with **AWS and Microsoft Azure** to perform business analytics, manage clusters, implement Machine Learning lifecycle to check the usage patterns of the data and check insight for data filtering and transformation.
- Optimized **Cloudera and Hortonworks Hadoop clusters**, tuning **YARN, HDFS, and MapReduce parameters**, reduce job execution time, and improve **cluster stability under high workloads**. Representing **scalable data warehouses** using **Erwin and Kimball methodologies** to integrate diverse data sources, optimizing **star and snowflake schemas**.
- Demonstrated **interactive dashboards** using **Tableau, Power BI, and Looker**, integrating with cloud-based data lakes, warehouses, and extracting data from various sources including relational database. Adapt to performance optimization, scalability and best practice for cloud based **BI solutions**.
- For seamless collaboration across teams and create comprehensive system architecture, data flow diagrams and API specifications **Confluence, SharePoint, and Lucid chart** applications are used. Proficient in **CSV, Parquet, ORC, Avro, JSON, XML and TXT** and design of unified ingestion frameworks to enable seamless data transformation across data storage.
- Skilled in making OLAP cubes using **Kyvos, Druid, AAS, and SSAS** to support multi-dimensional analysis implementing **columnar storage optimizations**, and pre-aggregated roll-ups for **sub-second query performance** on datasets exceeding **billions of rows**.
- Automated **end-to-end deployment** of data pipelines using **Jenkins, Docker, Kubernetes, and Terraform**. Implemented **robust CI/CD pipelines** for **version-controlled, test-driven** deployments using **Git, GitHub, GitLab, SVN, and CVS**.
- Proven ability to collaborate with **cross-functional teams**, mentor junior engineers, conduct **code reviews, architecture discussions**, and rapidly **adopt emerging technologies** to drive **continuous innovation and process optimization**.

TECHNICAL SKILLS

- **Big data Technologies:** PySpark, Hadoop, HDFS, Spark, Hive, Sqoop, Oozie, BigQuery, MapReduce, Kafka, Kinesis, Spark SQL, Core Spark, Cassandra, Snowflake, Avro, Airflow, DynamoDB, Cloudera.
- **Database tools:** Oracle SQL, NoSQL, Mongo DB, MySQL, CSV, Parquet, XML, PostgreSQL, SQL Server
- **Cloud Technologies: Azure**
- Azure Data Lake, Data factory, Azure Databricks, Azure SQL database, Azure SQL Datawarehouse, Azure Synapse Analytics, Azure Data Lake Storage, Azure Stream Analytics, Azure Cosmos DB, Azure Machine Learning, Azure Databricks, Azure PostgreSQL, Microsoft Fabric.

AWS

S3, EC2, REDSHIFT, REDIS, EMR, Kinesis, Lambda, Step Functions, Terraform, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch and IAM, EDS, Cloud Run, Cloud Functions

- **Programming languages and tools:** Python SQL, Spark UI, Visual Studio Code, Eclipse IDE, PyCharm, Jupyter Notebooks, Scala.
- **Web Programming:** HTML, CSS, Java Script, Angular 17, Django, RESTful APIs, Hibernate
- **Data Warehousing:** Snowflake, Redshift, Teradata.
- **Data visualization:** Power BI, Tableau & Looker.
- **Python Libraries:** Matplotlib, Pandas, NumPy, Scikit-learn, SciPy, PyTorch, TensorFlow
- **CI/CD & DevOps:** Git, JIRA, Crontab, Unix Sockets, Jenkins, Docker, Kubernetes, GitHub, CVS
- **Data Analysis & Processing:** Spark, Data Transformation, Feature Engineering, Model Training & Performance Evaluation

PROFESSIONAL EXPERIENCE:

Edward Jones

Azure Data Engineer

May 2024- Present

Responsibilities:

- Developed and programmed **SQL** queries in **Postgre SQL** to support data extraction, transformation and reporting across multiple databases and business units and use of **Python** libraries like **Pandas**, **NumPy**, **SQL Alchemy** to support reproducible data pipelines.
- Created and optimized **data pipelines** using **Azure Data Lake** and **Azure Data Factory** to engage linked datasets and services and automate data across platforms like **Azure Databricks**, **Azure SQL database**, **Azure data warehouse**, **Azure Synapse** to provide efficient cloud-based operations.
- Implement end-to-end data pipelines using **Microsoft Fabric**, **Lakehouse's** and **Notebooks** to support scalable analytics across **structured** and **semi-structured** data and use of **One Lake architecture** to centralize **data storage**, reduce duplication and enable cross-domain **collaboration** across **business units**.
- Extracted from **Hadoop** data storage and processing, **Apache Spark** for data analytics across various projects. Use of **Snowflake** and **Databricks** for cloud-based warehousing and big-data analytics for efficient data engineering solutions.
- Skilled in incorporating diverse data formats like **JSON**, **CSV** form different formats exploiting and using streaming technologies like **Kafka** for real time processing and migrating relational databases from **SQL server** to **Azure data warehouse**.
- Developed **Stored procedures** and batch jobs to automate **data extraction** from **IDMS** segments and integrate downstream analytics pipelines and building reusable data flows between **SQL Server** for third-party applications.
- Successful migration of data from on-premises to cloud platforms like **Snowflake** to ensure seamless, efficient, and scalable data operations. Integrated **Mongo DB** and aggregation to transform and analyze large datasets in distributed environment.
- Experience in use of **OLAP** tools like **Azure Analysis Services** and connecting them to **Power BI** to develop interactive dashboards to enable advanced analytics and intuitive visualizations.
- Development and optimization of **SQL server** and data solutions are performed using **Mongo DB** which includes schema design, indexing strategies and stored procedures for high-performance analytics and transactional systems. Use of **CDC** to stream updates from **Apache Spark** to **SQL server** for low latency data replication and analytics.
- Proficient in writing **Python** validation Scripts within **Databricks** for data quality checks and anomaly detection across datasets and integrate **Databricks** with **Azure Key Vault** for secure credential management and seamless data access across teams.
- Proficient in automating **Alteryx workflows** to streamline **ETL process** for finance reporting, risk aggregation and regulatory compliance integrating structured data from **Excel**, **Oracle** and **Salesforce** leveraging predictive modeling and data cleansing to reduce manual intervention in monthly close process and enable faster delivery of reconciled financial statements.
- Work on custom business applications using **Microsoft Power apps** to streamline data entry, approval workflows and reporting across finance and operations teams and integrate with **SQL server** to enable **real-time data access** and automate cross-platform process.

- Implement Collibra **Data Governance** Center to manage business glossaries, data dictionaries, and stewardship workflows across enterprise data domains.
- Designed and tested **RESTful APIs** using **Postman** for validating request/response payloads, headers, and **authentication** workflows across microservices.
- Implement **Data Quality** rules, profiling and cleansing using **Informatica Data Quality** to ensure accuracy, completeness and consistency across critical **business domains** and design **reusable mappings** and parameterized workflows to support **ingestion pipelines** and reduce development overhead across projects.
- Utilization of tools like **Apache Airflow**, **Azure Data Factory** for scheduling, automating, and arranging **complex data workflows** and optimizing **task dependencies** ensuring timely execution and streamline end-to-end data pipelines for efficient operations.
- Design and deploy custom business applications using **Microsoft Power apps** to streamline data entry, approval workflows and reporting across finance and operations teams, implement **role-based access** controls to embed **Power BI** dashboards within Power apps to deliver secure, interactive analytics experiences.
- Use of **Microsoft Purview** to automate data discovery, lineage tracking and classification across **Azure data lake**, **Synapse** and **Microsoft Fabric** to ensure **GDPR** and financial **compliance** by categorizing **PII** and transaction data, establish secure access policies and impact analysis across audit-readiness and regulatory reporting across enterprise.
- Integrate **machine learning models** into **ETL pipelines** using **Python**, **PySpark** and **Databricks** to automate data quality check, anomaly detection, and predictive analysis to enable real-time decision making and improve data pipeline efficiency.
- Use **GitHub** for version control and collaborating development, manage **code repositories**, implement **branch strategy** and integrating with **CI/CD** pipelines to ensure consistent, high-quality code deployment across data engineering and analytics projects.
- Experience in implementing **Agile methodologies** to enhance project efficiency using tools like **JIRA** for application lifecycle process, seamless collaboration, and on-time delivery.
- Created detailed **data flow** diagrams (DFDs), **ETL** architecture maps and system integration workflows using **Microsoft Visio** to support cross functional teams and document end-to-end pipeline architecture in **Visio** including source systems, layers and target data stores across **Azure** and **Snowflake**.

Environment: PostgreSQL, SQL, Pandas, NumPy, SQLAlchemy, Azure Data Lake, Azure Data Factory, Azure Databricks, Azure SQL Database, Azure Data Warehouse, Azure Synapse, Microsoft Fabric, Lakehouse, Notebooks, OneLake, Hadoop, Apache Spark, Snowflake, Databricks, JSON, CSV, Kafka, SQL Server, IDMS, MongoDB, Azure Analysis Services, Power BI, CDC, Azure Key Vault, Alteryx, Excel, Oracle, Salesforce, Microsoft Power Apps, Collibra, Postman, RESTful APIs, Informatica Data Quality, Apache Airflow, Microsoft Purview, PySpark, GitHub, Jenkins, Terraform, JIRA, Microsoft Visio.

Freddie Mac

Senior Data Engineer

January 2023- May 2024

Responsibilities:

- Optimized complex **SQL** queries for relational databases like **PostgreSQL**, **MySQL**, **Redshift**, and integrated **NoSQL** systems like **DynamoDB** and **MongoDB** for data solutions along with implementation of data architecture on **AWS S3** with partitioning, lifecycle policies to manage high-volume datasets efficiently.
- Designed and maintained **data pipelines** using **AWS services** including **S3**, **Redshift**, **Lambda**, **Glue** to support enterprise data warehousing, analytics and reporting and develop ETL workflows using Python and orchestration tools like Apache Airflow to automate ingestion, transformation and loading of structured and semi-structured data.
- Well-versed in use of big data technology tools like **Spark** to manage large-scale batch processing for distributed data storage and ensure the high performance of applications which include data transformation, window aggregation, and streaming analytics.
- Develop **RESTful API** using **Flask** and **Fast API** to serve real-time data services and integrate seamlessly with front-end and third-party applications.
- Use of Python Libraries like **Pandas** and **SQL Alchemy** to perform **data wrangling**, validation, and transformation across large datasets for analytics and **machine learning** workflows.
- Integrated Lake formation with **AWS Data Platform** accounts for centralized authentication and authorization management. Used **HashiCorp** vault in AWS services like **Secret manager**, **RDS** to deploy secure management solutions.
- Effective migration to **AWS Redshift** from on-premises to ensure zero downtime with minimal loss of data and use of **Teradata** for **Snowflake** migration to optimize storage and improve query execution time.
- **Cassandra** is used for storing semi-structured data as a **NoSQL** database for high availability, fault tolerance and overseeing high-throughput workloads for globally distributed applications.
- Deployed **PyTest** for data validation and integrity checks in **ETL** workflows and ensured accurate transformation logic by designing unit test for **ETL pipelines** and identify data inconsistencies, data accuracy, and data integrity across critical business operations.
- Followed robust **data governance** frameworks which help to follow data privacy regulations which improve data quality across multiple domains and enable development of **API** using **Python** and **SQL**.

- Implemented logging mechanisms and exception handling using **AWS CloudWatch** for efficient system debugging monitoring and robust application performance for diverse projects.
- Leverage use of **big-data policies** like **Hadoop** and **Apache Spark** to process distributed datasets and enable scalable analytics across batch and streaming environment and monitor **DAG** in **Apache Airflow** to orchestrate multi-step data workflows, manage dependencies, and ensure timely execution of data jobs.
- Utilize **Tableau** for making interactive dashboards and reports for making data-driven solutions, skilled in API to facilitate seamless data integration, exchange, retrieval, and processing using **Snowflake**.
- Skilled in using various file formats like **Parquet** to optimize data storage and retrieval processes which ensures high performance, efficient compression, compatibility with big-data processing frameworks for scalable and cost-effective solutions.
- Proficient implementation of robust access mechanisms using **IAM** policies to enforce secure and scalable permissions and creating custom logic for specific security, access requirements, and compliant data access in diverse environments.
- Experienced in deploying and managing containerized applications like **Kubernetes** for enabling automated scaling, rolling updates, service discovery, and seamless orchestration of microservices in cloud-native environments.
- Proficient utilization of tools like **Apache Airflow** to design, automate and deploy complex workflows. Skilled in managing task dependencies for optimizing pipeline performance and ensuring timely execution of data processes in scalable environments.
- Integrate **AWS services** with **CI/CD** pipeline using **GitHub** and **Jenkins** to automate deployment and maintain **version control** across data engineering projects.
- Collaborate with **cross-functional teams** to gather requirements, design data models, and deliver analytics -ready datasets for **business intelligence** and operational reporting. **Document data architecture, pipeline logic and operational workflows, audit readiness, and cross-functional transparency.**
- Expertise in implementing **Agile methodologies** which includes sprint planning, backlog grooming, and iterative development to improve project efficiency and adaptability. Skilled in the use of **Jira** for tracking Agile workflows and managing tasks.

Environment: PostgreSQL, MySQL, Redshift, DynamoDB, MongoDB, AWS S3, Lambda, Glue, Python, Apache Airflow, Spark, Flask, Fast API, Pandas, SQL Alchemy, Lake Formation, HashiCorp Vault, Secret Manager, RDS, Teradata, Snowflake, Cassandra, PyTest, SQL, AWS CloudWatch, Hadoop, Tableau, Parquet, IAM, Kubernetes, GitHub, Jenkins, Jira.

Instacart

Data Science Analyst

October 2020- January 2023

Responsibilities:

- Designed and implemented scalable data architecture for warehouses and data lakes with business requirements and best practices. Use of **Oracle SQL** to evaluate optimal data technologies for required project needs.
- Automated data transformation process to improve data quality and minimize manual intervention. Integration with **SQL** server for **ETL process** for extraction and transformation stages to enhance pipeline efficiency.
- Optimized data pipelines using **Spark-Scala** for handling large-scale data processing for analytics to integrate big data workflows for efficient and fast **SQL** based operations which helps to enable seamless data integration, analytics, and scalability for diverse business use cases.
- Integrated/Ingested data from various **APIs** using **Kafka** to ensure low latency data processing, effective synchronization, and interoperability across platforms for dynamic applications. Performed data aggregations to streamline consumption patterns for analytical and operational use case.
- Migration of data systems from **Informatics** to cloud-native solutions like **Terraform** and **Snowflake** to improve performance and incorporate tuning techniques to enhance query efficiency, reduce costs, and improve scalability. Also included MSSQL server read replicas to monitor systems and manage the infrastructure.
- Utilization of **Cassandra** for **NoSQL** solutions for high availability and reduce fault tolerance which results in efficient storage and retrieval of time-series and semi-structured data in distributed systems and enables flexible data modeling with microservices based applications.
- Efficient use of **data reconciliation** techniques to compare source and target-based datasets using **SQL-based validation** scripts, identify discrepancies and ensure data completeness. Use of **JSON** schema ensuring compatibility and consistency in **data ingestion** pipelines.
- Experience in use of **Kafka** logs to track real time data streaming problems and check reliability across distributed processing systems and manage missing or corrupt data preventing pipeline crashes and ensuring data integrity.
- Distinguished various patterns of data by using **Power BI** and integrating with **Snowflake** for optimized data retrieval and improve query performance with partitioning and caching strategies. Using **API** gateway to provide secure and scalable endpoints for downstream applications.
- Employed use of various file formats for storage purposes like **Hive** to enable faster query performance and efficient data retrieval in big data environments and supports schema evolution for effective serialization/deserialization.
- Automated **CI/CD pipelines** using **Jenkins** to monitor performance to ensure high availability and fault tolerance and combining ETL workflows for automated scaling and deployment. This also ensured quality assurance and a reduction of manual intervention for code integration.

- Designed and managed data flows using **Apache Workflow** for scheduling, automating and orchestrate **ETL pipelines** and optimized task dependencies and workflow performance to ensure timely execution of business-critical data processes.
- Incorporated successful workflows and code integrity using **GitHub** and **Git** to automatically build and deploy processes for faster delivery cycles; this also helped for using project management features to monitor progress and improve team productivity.
- Leveraged **Jira** for detailed reporting m backlog grooming and sprint reviews to enhance project visibility and improve efficiency and manage test cases, defect, and test execution to match quality standards throughout development cycle.

Environment: Oracle SQL, SQL Server, Spark-Scala, Kafka, Terraform, Snowflake, Cassandra, JSON Schema, Power BI, API Gateway, Hive, Jenkins, Apache, Apache, Apache Workflow, GitHub, Git, Jira, Informatica, MySQL.

Amazon

Data Engineer

December 2018- October 2020

Responsibilities:

- Created scalable data architecture for data warehouses and data lakes which ensures efficient storage, retrieval, and processing of large datasets using **Oracle SQL**. Optimization of data pipelines is implemented by developing ETL workflows using **Apache Airflow**.
- Optimized relational schemas using **Oracle, MySQL, DB2** and **PostgreSQL** for efficient data storage, indexing and retrieval, tuning of SQL queries and stored procedures is done using **Oracle SQL** for transactional processing, reporting and data transformation.
- Implemented real-time and batch **ETL** processing using **Informatica** to support real-time analytics and business intelligence reporting. Errors and logging mechanisms are overseen using **Informatica** to track job features, automate retries and ensure data integrity.
- Ingested real-time and batch data from various sources like **Kafka** for seamless data flow and low-latency processing and integrated with third-party data sources using **RESTful API, Flask** for secure and efficient data exchange for ingestion workflows.
- Complex queries are optimized using **Oracle SQL, SQL** server for improving execution time and resource utilization and implemented window functions to perform analytical computations over partitions for business insight. Developed advanced stored procedures, functions and triggers using **PL/SQL** for automating business logic for large datasets.
- Used **Git, GitHub** for version control and manage source code and track changes for collaborative development and managed repository workflows using pull requests, code reviews, and merge controls to maintain high-quality code and enforce development standards.
- Automated Continuous Integration and Continuous deployment using **Jenkins** to build, test and deploy the code for reliable delivery cycles and configured **Jenkin pipelines** with version control like **GitHub** and **Bitbucket** for source code management and trigger-based results.
- Experienced in managing **Hadoop** clusters for storing massive datasets and large volumes of unstructured data which enables scalable and distributed data processing. Leveraged Cloud tools like **AWS Glue** for data processing, transformation enhancing overall performance of big data workflows.
- Identified business requirements, data types, and workloads to determine suitable cloud services with minimal disruption during migration and use of **AWS Migration hub** to streamline and automate the migration process. Use of data replication and synchronization for real-time consistency between on-prem and cloud environment during migration process.

Environment: Oracle SQL, Apache Airflow, Oracle, MySQL, DB2, PostgreSQL, Informatica, Kafka, RESTful API, Flask, PL/SQL, Git, GitHub, Jenkins, Bitbucket, Hadoop, AWS glue, AWS migration hub.

EDUCATION:

University of Colorado, Denver

Master of Science in Information Systems, GPA-3.7